

7) practice problems:-Q#1:- Data:-

$$c = 100 \text{ km/hr}$$

part (a):-

$$d = 175 \text{ km}$$

end to end delay:-

Sol:-

$$d_{\text{prop}} = d/c = 175 \text{ km} / 100 \text{ km/hr} = 1.75 \text{ hr} \quad \text{Ans.}$$

part (b):- 8 cars.Sol:-

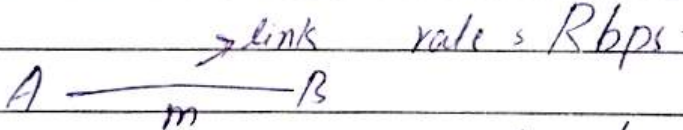
$$\text{Transmission delay} = d = L/R$$

$$12 \text{ bits} \times 8 = 96 \text{ bits}$$

$$\text{propagation delay} = d/c = 175/100 = 1.75 \text{ hr}$$

$$1.75 \times 3600 = 6300$$

$$d_{\text{total}} = 6300 + 96 = 6396 / 3600 = 1.776 \text{ min}$$

Q#2:-

$$s = m/s$$

L bits.

- propagation delay  $s = m/m/s = \text{sec}$  Ans.
- transmission-time  $s = d_{\text{trans}} = L/R = \text{bits/bps}$  Ans.
- $d_{\text{total}} = d_{\text{trans}} + d_{\text{prop}}$  Ans.
- At  $t = d_{\text{trans}}$  the last bit is onto the link.

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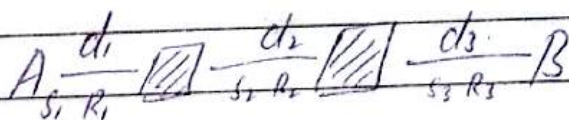
- (c)  $d_{prop} > d_{trans}$  then the first bit hasn't reached B.  
 (f)  $d_{prop} \approx d_{trans}$  then the first bit is on its destination (Host B).  
 (g)  $s = 2.5 \times 10^8 \text{ m/s}$   
 $L = 1500 \text{ bytes}$   
 $R = 10 \text{ Mbps}$   
 $d = ?$

~~$\frac{L}{R} = d$~~  (i)  ~~$\frac{L}{R} = d$~~  (c)  ~~$\frac{12000 \times 2.5 \times 10^8}{10 \times 10^6}$~~  (c)  ~~$\frac{12 \times 10^3 \times 2.5 \times 10^8}{10 \times 10^6}$~~

$d_{trans} = \frac{L}{R} = \frac{12000 \text{ bits}}{10^7} = 0.0012 \text{ sec}$   $\approx 30 \times 10^{-4}$   
 $\approx 3 \times 10^{-3}$

$d_{prop} \text{ m/s} = 0.0012 \text{ sec}$   
 $\approx 0.0012 \times 2.5 \times 10^8$   
 $\approx 1.2 \times 10^{-3} \times 2.5 \times 10^8$   
 $\approx \frac{3 \times 10^5 \text{ m}}{10^3} \text{ i.e. } 300 \text{ km or } 300000 \text{ m Ans.}$

Q43.



$d_{trans} = \frac{L}{R_1} + \frac{L}{R_2} + \frac{L}{R_3}$

$d_{nuctal} = \frac{L_1}{R_1} + \frac{L}{R_2} + \frac{L}{R_3} + \frac{d_1}{s_1} + \frac{d_2}{s_2} + \frac{d_3}{s_3} + d_{dproc}$

$d_{prop} = \frac{d_1}{s_1} + \frac{d_2}{s_2} + \frac{d_3}{s_3}$

$d_{proc} = d_{dproc}$



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Q#4:  $L = 1500 \text{ bytes}$   $\Rightarrow 1500 \times 8 = 6000 \text{ bits}$   $\times 2 = 12000$   
 $R = 2.5 \text{ Mbps}$   $\Rightarrow 2.5 \times 10^6 \text{ bps}$   
 delay = ?

Sol:

$T = \frac{L}{R}$  when intensity of traffic known.

$$\text{delay} = \frac{L}{R} = \frac{6000 \times 2}{2.5 \times 10^6} = \frac{6 \times 10^3 \times 2}{2.5 \times 10^6} = \frac{2.4 \times 10^3}{0.0025} = 0.0048 \text{ sec}$$

4.8ms  $\rightarrow$  for one packet.

$$4.8 \times 4 + 2.4 = 19.2 + 2.4 = 21.6 \text{ ms}$$

Q#5:  $N$ -packets Avg queueing-delay?  
 (a)  $L$  = length Sol:  
 $R$  = Transmission rate.

packet 1: 0

$$\text{packet 2} = \frac{L}{R}$$

$$\text{packet 3} = \frac{2L}{R}$$

$$\text{packet } N = \frac{(N-1)L}{R}$$

$$(b) \frac{NL}{R} \text{ seconds}$$

Sol:

$$\text{packet } 1 = \frac{NL}{R}$$

$$\frac{L}{R} \frac{N(N-1)}{2N} = \frac{L(N-1)}{2R}$$

Ans

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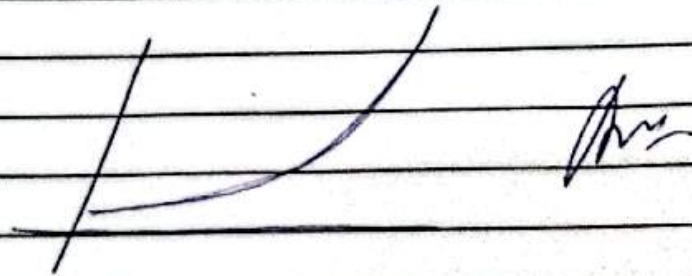
Q#6: (a)  $I = \frac{\lambda \cdot a}{R}$   $d_{trans} = \frac{IL}{R(1-I)}$

$$d_{trans} = \frac{L}{R}$$

$$d_{total} = \frac{IL}{R(1-I)} + \frac{L}{R} \Rightarrow \frac{L}{R} \left( \frac{I}{1-I} + 1 \right) \Rightarrow \frac{I + 1 - I}{1-I} = \frac{1}{1-I}$$

$$d_{total} = \frac{L}{R} \left( \frac{1}{1-I} \right) \text{ ms}$$

part (b) s)



Q#7: Throughput: data-transferred  
time

$$\text{Throughput} \leq R_1 \text{ or } R_2$$

$$\text{Throughput} \leq R_2 \text{ or } R_1$$

$$\text{Throughput} \leq R/M$$

$$\text{Throughput} \leq \min(R_1, R_2, R/M)$$

ms



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Q#8.

Loss-probability =  $p$ .

Success-probability for each one =  $1-p$ .

as there are three so

$$(1-p)(1-p)(1-p)$$

$$= (1-p)^3 \text{ Ans.}$$

Q#9.

1TB =  $10^{12}$  bytes.

50TB =  $50 \times 10^{12}$  bytes.

$$50 \times 10^{12} \times 8 = 4 \times 10^{14} \text{ bits.}$$

$$\text{trans} = \frac{4 \times 10^{14}}{10 \times 10^6} = 4 \times 10^6 \text{ seconds.}$$

$$\frac{40 \times 10^6 \text{ seconds}}{60 \times 60 \times 24} = 46.3 \text{ days}$$

2) I will prefer faster-overnight delivery



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Q#10:-

$L = 4000 \text{ bits}$

$R = 10 \text{ Mbps}$

(a)  $d_{\text{trans}} = ?$

(b) traffic intensity: ?

Sol:-

(a)  $d_{\text{trans}} = \frac{L}{R} = \frac{4 \times 10^3}{10 \times 10^6} = \frac{4 \times 10^3}{10^7} = 4 \times 10^{-4} \text{ seconds} \text{ Ans}$

(b) Traffic intensity  $= \frac{L}{R} \times \text{average packet arrival rate}$

$= \frac{4 \times 10^{-4} \times 30}{1} = 120 \times 10^{-4} \text{ Ans}$

Q#11:-

$R_1 = 100 \text{ Mbps}$

$R_2 = 10 \text{ Mbps}$

$R_3 = 1 \text{ Mbps}$

$L = 18000 \text{ bits}$

$L = 3 \text{ km (distance)}$

Sol:-

(a)  $d_{\text{trans}} = \frac{18000}{10 \times 10^6} = \frac{18 \times 10^3}{10^7} = 18 \times 10^{-4} \text{ Ans}$

(a)  $d_{\text{trans}}$  of  $R_2$

(b)  $d_{\text{prop}}$  of  $R_1$

(c) Source of packet delay

(d) Datagram move?

$d_{\text{prop}}$

(b)  $c/s = \frac{3 \times 10^3}{3 \times 10^8} = 1 \times 10^{-5} \text{ Ans}$

(c) (i) processing delay

(ii) propagation delay

(iii) Transmission delay

(iv) Queuing delay

(d) Network layer

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1 Gbps, 1000 Mbps

Date: 2/9/26

Q#12- Router 1  $\rightarrow$  100 Mbps, same propagation delay.  
Router 1  $\rightarrow$  Router 2  $\rightarrow$  25 Mbps, 200 ms propagation delay.  
Router 2  $\rightarrow$  Client  $\rightarrow$  1 Gbps, 0 propagation delay.  
 $L = 20000$  bits.

No processing & queuing delay.  
Q#12- (a) End-to-End delay?

$$d_{trans} = \frac{20000}{100 \times 10^6} = \frac{2 \times 10^4}{1 \times 10^8} = 2 \times 10^{-4} \text{ seconds} = 0.2 \text{ ms}$$

$$d_{prop} = 200 \text{ ms} \quad d_{total} = 200 + 0.2 = 200.2 \text{ ms}$$

$$d_{trans} = \frac{20000}{25 \times 10^6} = \frac{2 \times 10^4}{25 \times 10^6} = 0.08 \times 10^{-2} \times 10^3 = 0.8 \text{ ms}$$

$$d_{prop} = 200 \text{ ms}$$

$$d_{total} = 200 + 0.8 = 200.8 \text{ ms}$$

$$d_{trans} = \frac{20000}{1000 \times 10^6} = \frac{2 \times 10^4}{10^9} = 2 \times 10^{-5} \times 10^3 = 2 \times 10^{-2} = 0.02 \text{ ms}$$

$$d_{prop} = 0$$

$$\text{Total delay} = 0.2 + 200.8 + 0.02 = 201.02 \text{ ms}$$

(b) throughput  $\leq \min(100, 25, 1000)$   
 $= 25 \text{ Mbps}$



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Q#13:  $L = 2500 \times 8 = 20,000 \text{ bits}$

(a) Bottleneck-link:

$$\min(100, 100, 25, 10, 100) \\ = 25 \text{ mbps (Bottleneck)}$$

c) it limits the max throughput. *Ans*

(b) If traffic on bottleneck increases then queuing delay increases.

(c) if the bottleneck is upgraded then this will increase the throughput (end-to-end). *Ans*

(d)  $d_{\text{trans}} = \frac{20000}{100 \times 10^6} = 2 \times 10^{-4} \times 10^3 = 0.2 \text{ ms}$

$d_{\text{prop}} = 15 \text{ ms}$

$d_{\text{e}} = 15 \text{ ms} + 0.2 \text{ ms} = 15.2 \text{ ms}$

$d_{\text{trans}} = 20000 / 100 \times 10^6 = 0.2 \text{ ms}$

$d_{\text{prop}} = 20 \text{ ms} \Rightarrow 20.2 \text{ ms}$

$d_{\text{trans}} = 20000 / 25 \times 10^6 = 0.8 \text{ ms}$

$d_{\text{prop}} = 40 \text{ ms} \Rightarrow 40.8 \text{ ms}$

$d_{\text{trans}} = 20000 / 10 \times 10^6 = 2 \text{ ms}$

$d_{\text{prop}} = 30 \text{ ms} \Rightarrow 32 \text{ ms}$

$d_{\text{trans}} = 20000 / 10 \times 10^6 = 2 \text{ ms}$

$d_{\text{prop}} = 10 \text{ ms} \Rightarrow 12 \text{ ms}$

total delay  $= 15.2 + 20.2 + 40.8 + 32 + 12$   
 $= 120.2 \text{ ms}$

*Ans*